

DISTRIBUTIONAL GROWTH ACCOUNTING: EDUCATION AND THE REDUCTION OF GLOBAL POVERTY, 1980–2019*

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This article quantifies the role played by education in the reduction of global poverty. I propose tools for identifying the contribution of schooling to economic growth by income group, integrating imperfect substitution between skill groups into a macroeconomic growth decomposition. I bring this “distributional growth accounting” framework to the data by exploiting a new microdatabase representative of nearly all of the world’s population, new estimates of the private returns to schooling, and historical income distribution statistics. Education can account for about 45% of global economic growth and 60% of pretax income growth among the world’s poorest 20% from 1980 to 2019. A significant fraction of these gains was made possible by skill-biased technical change amplifying the returns to education. Because they ignore the distributional effects of schooling, standard growth accounting methods substantially underestimate economic benefits of education for the global poor. *JEL codes:* D31, O11, I24, I25, I26.

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I. INTRODUCTION

The past decades witnessed major improvements in access to public services in developing countries. These improvements were evident in progress made on indicators as varied as school enrollment, health care coverage, and access to drinkable water ([United Nations 2023](#)).

How useful these policies have been at generating income growth for the global poor remains an open question. Education in particular has expanded massively, yet its contribution to global poverty reduction remains unclear. The economic effects of education depend on who has access to schooling, the returns to schooling, and general equilibrium effects. Because of difficulties in quantifying these channels, we lack estimates of the benefits of schooling for the global poor. Addressing this gap is of fundamental importance for policy, in a world where the majority of children from low-income households are enrolled in public schools.

This article estimates the aggregate and distributional effects of worldwide educational expansion from 1980 to 2019. Leveraging a new microdatabase on individual incomes, a “distributional growth accounting” model of education and the wage structure, and complementary quasi-experimental evidence, I quantify the contribution of schooling to growth for different groups of the world distribution of income. I find that education accounts for a substantial fraction of global economic growth and extreme poverty reduction during this period, placing education policies at the forefront of economic progress.

The starting point is a new microdatabase representative of 97% of the world’s population based on household surveys fielded in 154 countries around 2019, collected from data repositories and country-specific sources. Each survey provides information on individual incomes, education, age, gender, and other socioeconomic variables. I harmonize each survey to construct a single data set covering 10 million surveyed individuals, yielding a comprehensive microdatabase to analyze the contemporary structure of global poverty and inequality.

Starting with this new data set, I develop methods to estimate what the world distribution of income would have looked like in 2019 had schooling not improved since 1980. I combine standard growth accounting tools with a model of education and the wage structure à la [Goldin and Katz \(2007\)](#). In this distributional growth accounting framework, expanding education raises

aggregate labor income by the private return to schooling, which is endogenous to relative supply (education) and relative demand (the skill bias of technology) for skilled labor. Schooling also reduces inequality by lowering the relative wage of skilled workers as their supply increases.

The model delivers three main results. First, the contribution of education to growth is larger than in a standard growth accounting decomposition. The return to schooling would be higher today had education not expanded, because the supply of skilled workers would be lower. As a result, reverting education to its 1980 levels would imply a greater loss in output than what 2019 returns to schooling suggest. Second, education disproportionately benefits low-income earners: inequality would be higher had the relative supply of skilled workers not increased. Third, the total effect of education on growth can be decomposed into two forces: the “independent” effect that education would have had if labor demand had not changed, and skill-biased technical change, which raises the returns to schooling by increasing relative demand for skilled labor.

I begin by estimating the total contribution of education to global poverty reduction. I construct a counterfactual world distribution of income in three steps. First, I downgrade education levels in each survey until they match the 1980 distribution of educational attainment. Second, I reduce individual incomes accordingly, combining new country-specific estimates of returns to primary, secondary, and tertiary education with supply effects derived from the model. Third, I compare the resulting counterfactual to the actual evolution of incomes, yielding an estimate of the contribution of education to growth for different groups in each country.

I find that private returns to schooling account for about 45% of global economic growth (40%–65% depending on the specification) and 60% (50%–100%) of pretax income gains among the world’s poorest 20% over 1980–2019 (see [Figure I](#)). Education also explains about one-third of the reduction in the share of the world’s population living in extreme poverty.

A natural concern with this analysis is that private returns to schooling may differ from the aggregate effects of education on growth. This framework may also miss important channels through which education affects the income distribution. To address these limitations, I validate my approach with new quasi-experimental evidence by extending existing work on three

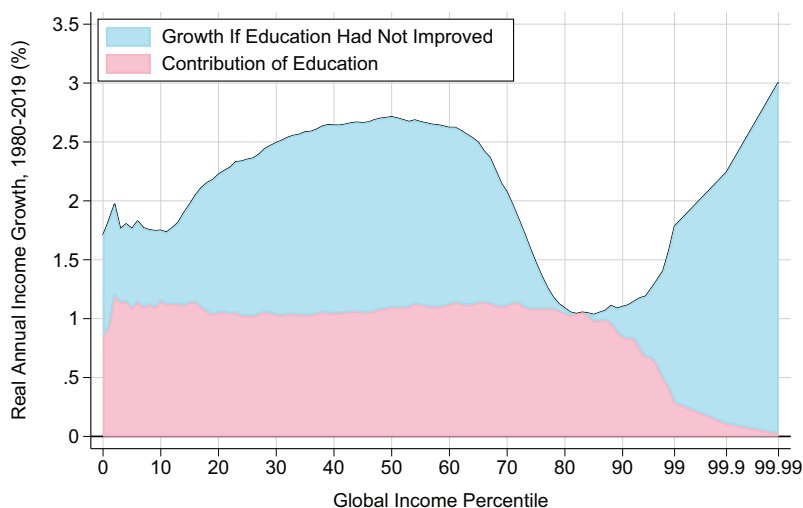


FIGURE I

Education and the Distribution of Global Economic Growth, 1980–2019

The figure plots total real income growth by global income percentile from 1980 to 2019, decomposing it into a part that can be explained by education and an unexplained component. The upper shaded area represents the growth rates that would have prevailed absent any improvement in educational attainment of the world's working-age population during this period. The lower shaded area represents the corresponding contribution of education to economic growth. From 1980 to 2019, the average income of the 20th percentile of the world distribution of income grew by 2.2% a year, 1 percentage point of which can be explained by education. Education thus accounts for about 45% of growth among this group. See [Online Appendix B](#) for additional methodological details.

large-scale schooling initiatives in India ([Khanna 2023](#)), Indonesia ([Duflo 2001](#)), and the United States ([Acemoglu and Angrist 2000](#)). Combining microdata on the distribution of income with differential exposure to each program across subnational regions, I document two facts. First, educational expansion had large causal effects on aggregate regional incomes, comparable to individual returns found in the same contexts. Second, all three policies disproportionately benefited low-income earners. The distributional growth accounting framework accurately replicates these findings, suggesting that it provides a good methodological foundation to study the role of education in shaping the distribution of income growth.

More generally, my estimates should be considered conservative. My main specification relies on standard Mincerian returns,

which are typically lower than causal estimates, as I show in a new meta-analysis of 62 papers. I assume that education only affects labor income, excluding positive effects on physical capital and human capital externalities, on which there is now significant empirical evidence (e.g., [Moretti 2004](#); [Gennaioli et al. 2013](#); [Guo, Roys, and Seshadri 2024](#); [Queiró 2022](#)). My findings depend on three sets of parameters: the private returns to schooling, the degree of imperfect substitutability between skill groups, and the extent to which other production factors adjust. With plausible bounds for these parameters, education is found to account for 50%–100% of growth among the world's poorest 20%.

Methodologically, accounting for distributional effects of schooling within countries appears essential. Standard growth accounting, as in [Barro and Lee \(2015\)](#), typically combines cross-country data on years of schooling with a constant 10% return to conduct the same exercise as in this article. This approach could provide a good approximation. I find that it does not: education accounts for 16% of growth among the world's poorest 20% with this method, compared with 58% in my benchmark specification.¹ One reason is that global poverty cannot be measured accurately from cross-country data: the poorest individuals in the world do not all live in the poorest countries. The global poor also rely more on labor income and thus benefit more from education. Most important, standard growth accounting does not account for supply effects: had education not improved, incomes would be lower than what 2019 returns to schooling imply, and that of low-skilled workers disproportionately so. The contribution of this article lies in the use of new microdata and tools to quantify each of these channels.

I conclude by analyzing the role of skill-biased technical change in amplifying the returns to schooling, exploiting additional surveys fielded around 2000 in 109 countries representing 80% of the world's population. In the average country, about 30% of the benefits of education from 2000 to 2019 were made possible by skill-biased technical change. There is substantial

1. The standard approach also underestimates the contribution of education to aggregate growth by about 30%, for two main reasons. First, it implicitly assumes that the return to schooling only applies to a fraction of mixed income, while I provide evidence that mixed income is affected by education just as much as wages. Second, I account for imperfect substitution, which increases the contribution of schooling by magnifying losses from not expanding education.

heterogeneity across countries, however. Demand for skilled labor increased rapidly in high-income countries while it stagnated in the developing world. Taking stock of this evidence, I estimate that skill-biased technical change can explain up to 30% of aggregate economic growth, but less than 5% of growth among the world's poorest 20% during this period.

A large literature in labor economics uses the canonical labor supply-and-demand framework to relate changes in the wage distribution to educational expansion.² Concurrently, a considerable literature in macroeconomics investigates the contribution of human capital to economic development.³ These two methodological perspectives, one focused on within-country inequality and the other on cross-country dynamics, have remained relatively independent. The main contribution of this study is to unify them into a distributional growth accounting framework, which I use to quantify the role played by education in the reduction of global poverty.

This article also contributes to our understanding of the forces shaping the long-run evolution of the world distribution of income. Global inequalities have undergone major transformations in recent decades, including rapidly declining poverty and cross-country income convergence (Sala-i-Martin 2006; Chen and Ravallion 2010; Pinkovskiy and Sala-i-Martin 2016; Hammar and Waldenström 2020), the emergence of a new “global median class” (Lakner and Milanovic 2016), and skyrocketing top income inequality (Chancel and Piketty 2021). Among the many drivers shaping these dynamics, I isolate the contribution of education.

Finally, this work relates to an extensive empirical literature on the economic effects of education. Many studies document positive effects of schooling on individual earnings (Card 2001; Psacharopoulos and Patrinos 2004; Deming 2022). A more limited number of studies examine the general equilibrium effects of

2. This framework has been extensively applied to study wage inequality in the United States (e.g., Katz and Murphy 1992; Goldin and Katz 2007, 2008; Murphy and Topel 2016; Autor, Goldin, and Katz 2020). A growing literature extends this analysis to low- and middle-income countries (Fernández and Messina 2018; Vu and Vu-Thanh 2022; Khanna 2023).

3. The recent literature focuses on development accounting (e.g., Hall and Jones 1999; Hsieh and Klenow 2010; Gennaioli et al. 2013; Jones 2014; Hendricks and Schoellman 2018). Growth accounting dates back to Solow (1957), although worldwide perspectives are more recent (Mankiw, Romer, and Weil 1992; Barro and Lee 2015; Ahmed et al. 2020; Collin and Weil 2020).

education policies (e.g., [Duflo 2004](#); [Porzio, Rossi, and Santangelo 2022](#); [Hsiao 2023](#); [Khanna 2023](#)). I complement this evidence by revisiting the impact of education on growth and inequality in three different contexts. I also draw extensively on the literature to calibrate the parameters guiding my results, such as elasticities of substitution and returns to schooling, in line with recent efforts at bridging the micro-macro gap in the study of development ([Buera, Kaboski, and Townsend 2023](#)).

The rest of the article is organized as follows. [Section II](#) outlines the conceptual framework. [Section III](#) presents the data and methodology. [Section IV](#) describes the main results. [Section V](#) studies the role of skill-biased technical change in amplifying the benefits of education. [Section VI](#) concludes.

II. DISTRIBUTIONAL GROWTH ACCOUNTING

This section develops the distributional growth accounting framework. [Section II.A](#) presents the model specification. [Sections II.B](#) and [II.C](#) discuss how to estimate the contribution of education to aggregate growth and its distribution within countries in this model.

II.A. Model Specification

Output per capita at time t is produced from per capita physical capital K^t and labor H^t :

$$(1) \quad Y^t = F(K^t, H^t) = K^{t\alpha} (z^t H^t)^{1-\alpha},$$

where z^t denotes total factor productivity. Labor input per capita is a constant elasticity of substitution (CES) aggregator of the form:

$$(2) \quad H^t(\theta^t, L^t) = \left(\theta_1^t L_1^{\frac{\eta-1}{\eta}} + \theta_2^t L_2^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}.$$

L_1^t and L_2^t are the shares of low-skilled and high-skilled workers. θ_1 and θ_2 are factor-augmenting technology terms. η is the elasticity of substitution between low-skilled and high-skilled workers, assumed to be greater than one. Taking logs, the change in output per capita between two time periods can be decomposed into the contributions of physical capital, total factor productivity, and human capital:

$$(3) \quad \Delta \ln Y = \alpha \Delta \ln(K) + (1 - \alpha) \Delta \ln(z) + (1 - \alpha) \Delta \ln(H).$$

The share of output growth accounted for by each input is given by:

$$(4) \quad 1 = \underbrace{\frac{\alpha \Delta \ln(K)}{\Delta \ln Y}}_{\text{Physical Capital}} + \underbrace{\frac{(1 - \alpha) \Delta \ln(z)}{\Delta \ln Y}}_{\text{TFP}} + \underbrace{\frac{(1 - \alpha) \Delta \ln(H)}{\Delta \ln Y}}_{\text{Education + Skill Bias}}.$$

II.B. Aggregate Effect of Education

In [equation \(4\)](#), the change in the labor aggregator $\Delta \ln(H)$ results from the combination of two nonadditive forces: changes in relative demand for skilled workers θ (skill-biased technical change) and changes in their relative supply L (educational expansion). Accordingly, there are several ways of evaluating the contribution of education to growth, corresponding to different counterfactual questions.

1. *Total Versus Independent Effects of Education.* A first consideration relates to the level of technology θ at which the contribution of education is evaluated. As highlighted by [Caselli and Ciccone \(2019\)](#) and [Hendricks and Schoellman \(2023\)](#), the effect of education is not uniquely determined when skilled labor and the skill bias of technology are complements: it depends on the reference technology level at which it is evaluated. In our context, one option is to estimate the contribution of education to growth given the skill bias of technology observed in 2019:

$$(5) \quad \text{Share}_L^{\theta^{2019}} = (1 - \alpha) \frac{\ln H(\theta^{2019}, L^{2019}) - \ln H(\theta^{2019}, L^{1980})}{\ln Y^{2019} - \ln Y^{1980}}.$$

I refer to this as the “total” contribution of education, corresponding to the following counterfactual question: how much lower would growth have been if education had not improved, but skill-biased technical change had evolved the way it has? Alternatively, one may evaluate the effect of education given the technological skill bias observed in 1980:

$$(6) \quad \text{Share}_L^{\theta^{1980}} = (1 - \alpha) \frac{\ln H(\theta^{1980}, L^{2019}) - \ln H(\theta^{1980}, L^{1980})}{\ln Y^{2019} - \ln Y^{1980}}.$$

I refer to this as the “independent” contribution of education. The corresponding counterfactual question is: what would have been the contribution of education to growth absent skill-biased technical change? In the presence of technological change, we should expect demand for skilled labor to grow over time: $\Delta \theta_2 > \Delta \theta_1$. As a result, the total effect of education is typically greater than

its independent effect: the quantitative importance of education is maximized when evaluated at $\theta = \theta^{2019}$ and minimized when evaluated at $\theta = \theta^{1980}$. The difference between the two estimates captures an interaction term between education and technology, which reflects the role played by labor demand in amplifying the returns to schooling.

2. Short-Run Versus Long-Run Elasticities. A second consideration relates to how education and technology affect each other. The previous decomposition identifies the effect of education holding technology fixed. In practice, labor supply and labor demand may endogenously respond to each other.

On the one hand, labor demand may respond to educational expansion. I consider the endogenous technological choice model of [Hendricks and Schoellman \(2023\)](#). Formally, let firms choose the optimal skill weights θ_1^t and θ_2^t under the constraint of a technology frontier, as in [Caselli and Coleman \(2006\)](#):

$$(7) \quad (\kappa_1^t \theta_1^{t\omega} + \kappa_2^t \theta_2^{t\omega})^{\frac{1}{\omega}} \leq 1.$$

We assume that $\omega - \frac{\eta-1}{\eta} - \omega \frac{\eta-1}{\eta} > 0$. [Hendricks and Schoellman \(2023\)](#) show that solving for the firm's optimal skill bias yields a reduced-form aggregator of the form:

$$(8) \quad H^t(A^t, L^t) = \left(A_1^t L_1^{t \frac{\sigma-1}{\sigma}} + A_2^t L_2^{t \frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}},$$

where $\sigma > \eta$ and the technology parameters A_j are exogenous. In other words, allowing firms to choose their technological mix is theoretically equivalent to increasing the elasticity of substitution while holding the skill bias of technology fixed. η is a short-run elasticity of substitution, corresponding to the degree of imperfect substitutability observed when firms take technology as given. σ is a long-run elasticity, reflecting the fact that firms adjust their labor demand as a response to educational expansion. Using a long-run elasticity is thus equivalent to modeling technical change as indirectly resulting from education.

On the other hand, educational choices themselves may be shaped by skill-biased technical change. A large literature documents the role played by the economic environment in shaping schooling decisions (e.g., [Edmonds, Pavenik, and Topalova 2010](#); [Restuccia and Vandenbroucke 2013](#); [Atkin 2016](#)). Unfortunately, quantifying this channel is not possible in the absence of surveys fielded in the 1980s, which would be necessary to estimate the

degree to which schooling responds to labor demand.⁴ Throughout the article, I treat education as exogenous and leave the study of the determinants of schooling for future research.

3. *Interpretation and Comparison with Standard Growth Accounting.* How do these methodological choices affect growth accounting estimates and why? In the standard growth accounting exercise, human capital is a function of a fixed return to schooling, such as 10% per year of education. With imperfect substitution, in contrast, the return to schooling is endogenous:

$$(9) \quad \ln \left(\frac{w_2^t}{w_1^t} \right) = \ln \left(\frac{A_2^t}{A_1^t} \right) - \frac{1}{\sigma} \ln \left(\frac{L_2^t}{L_1^t} \right).$$

The return to schooling is increasing in relative demand for skilled workers (the skill bias of technology) and decreasing in their relative supply (education). This relationship has three implications.

First, the total contribution of education is greater than in the standard exercise. The intuition is the following: the return to schooling would be higher today if education had not improved, because the relative supply of skilled workers would be lower. As a result, the decline in output that would arise from bringing education back to its 1980 level is larger than what returns to schooling observed today suggest.⁵

Second, this contribution is maximized at lower values of the elasticity of substitution. Using a lower elasticity is equivalent to assuming that firms would not adjust their technological mix if education came back to its 1980 level, which generates a much larger decline in output in the counterfactual.

Third, the difference between the independent and total contributions of education is lower for higher values of the elasticity of substitution. Intuitively, choosing a higher elasticity amounts to interpreting skill-biased technical change as resulting from educational expansion, which limits the role played by exogenous technological change in shaping the returns to schooling.

4. In Section V, I provide evidence that skill-biased technical change has been limited in developing countries since 2000, however, which suggests that this channel has been modest at least during this period.

5. In practice, accounting for imperfect substitution is equivalent to using a return to schooling that falls in between the 2019 return $r(A^{2019}, L^{2019})$ and the return that would prevail without educational expansion $r(A^{2019}, L^{1980})$.

Conversely, using a lower elasticity implies that the total effect of education is larger, but also that a larger fraction of this total effect results from an interaction term between education and technology.

4. *Roadmap.* In summary, I consider two sets of specifications depending on (i) the level of technology at which the effect of education is evaluated and (ii) whether skill-biased technical change is interpreted as resulting from educational expansion (long-run elasticity) or not (short-run elasticity). [Section IV](#) starts by presenting results on the total effect of education, that is, the effect of education evaluated at $A = A^{2019}$. This counterfactual can be precisely estimated for the world as a whole, since it only requires data on education and earnings in 2019. I turn to the independent contribution of education in [Section V](#). This second exercise requires estimating skill-biased technical change, which necessitates historical surveys fielded at the beginning of the period of interest. Such surveys do not exist for the 1980s, but I was able to mobilize additional sources for a more restricted sample of countries since 2000.

II.C. Distributional Effects of Education

I now turn to the effect of education on inequality. Consider an increase in the share of skilled workers from L_2^{1980} to L_2^{2019} . Using [equation \(9\)](#), it is clear that:

$$(10) \quad \begin{aligned} & \ln \left[\frac{w_2(A^t, L^{1980})}{w_1(A^t, L^{1980})} \right] - \ln \left[\frac{w_2(A^t, L^{2019})}{w_1(A^t, L^{2019})} \right] \\ &= -\frac{1}{\sigma} \left[\ln \left(\frac{L_2^{1980}}{L_1^{1980}} \right) - \ln \left(\frac{L_2^{2019}}{L_1^{2019}} \right) \right] > 0. \end{aligned}$$

The relative wage of skilled workers would be higher in 2019 if education had not improved, because their relative supply would be lower. In other words, education reduces inequality. This effect is less pronounced for higher values of σ , reflecting a lower sensitivity of relative wages to relative supplies. As shown in [equation \(10\)](#), this result is independent of the level of the skill bias of technology at which it is estimated: it applies to both the independent and total effects of education.

III. DATA AND METHODOLOGY

This section presents the methodology used to estimate the total contribution of education to global poverty reduction from 1980 to 2019. [Section III.A](#) covers data sources. [Section III.B](#) turns to the estimation of the model. [Section III.C](#) validates the methodology with new evidence from three education policies.

III.A. Data Sources

1. *Survey Microdata.* The starting point is a data set of household surveys covering the joint distribution of personal income and education in 154 countries around 2019, which I have assembled for this study. These surveys come from two main sources (see [Online Appendix E](#) for more details).

The first data source is the International Labour Organization's database of household surveys ([ILO 2023](#)). Based on a considerable data collection effort and with the collaboration of statistical institutes, ILOSTAT harmonized a large number of surveys covering most countries in the world. I was able to access data from 1,300 surveys covering 130 countries over the 1990–2022 period. The database records individual-level information on wages, self-employment income, education, and other sociodemographic variables. Surveys are nationally representative and generally have large sample sizes. About two-thirds are labor force surveys designed to collect information on employment and earnings. About one-third are multipurpose surveys that record data on both labor market variables and other dimensions of households' conditions. I keep the survey conducted closest to 2019 in each country.

The coverage of the ILO microdata is remarkable, but the information on education is limited in some countries. A number of countries are missing, including big countries such as China and Russia. To expand the coverage and quality of the database, I turn to the websites of national statistical institutes and other sources, from which I download additional household surveys for 59 countries and harmonize them in the same way as the ILO.

[Table I](#) provides descriptive statistics on the resulting database. The data cover about 10 million individuals surveyed in 154 countries. These surveys are representative of over 95% of the population of each world region. The exception is the Middle East and North Africa, where surveys either do not exist or are inaccessible to researchers ([Ekhatior-Mobayode and Hoogeveen 2022](#)).

TABLE I
A NEW SURVEY MICRODATABASE ON EDUCATION AND INEQUALITY

	Countries	Observations	Share of population covered (%)
Europe	39	785,036	100
Northern America	2	539,862	100
Latin America	25	4,139,887	98.2
Asia-Pacific	30	2,773,236	97.9
Middle East and North Africa	15	936,170	81.0
Sub-Saharan Africa	43	892,362	99.0
World	154	10,066,553	97.2

Notes. The table reports the number of countries covered by the survey microdata, the total number of observations, and the share of the total population covered by world region and in the world as a whole. See [Online Appendix E](#) for more details.

Overall, the microdata cover about 97% of the world's population and 95% of the world's GDP.⁶

2. Returns to Schooling: OLS Returns. The second input for the analysis is a measure of the returns to schooling. I consider two options: returns to schooling estimated by OLS with the survey data or causally identified returns available from the existing literature.

In the main analysis, I rely on OLS estimates of returns to schooling by education level. The microdatabase reports information on four educational attainment categories: no schooling, incomplete or completed primary education, incomplete or completed secondary education, and incomplete or completed tertiary education. I estimate returns to schooling by level using modified Mincerian equations:

$$(11) \quad \ln y_i = \alpha + \beta_{pri} D_{i,pri} + \beta_{sec} D_{i,sec} + \beta_{ter} D_{i,ter} + X_i \beta + \varepsilon_i,$$

where y_i is total annual earned income from all jobs of individual i ; $D_{i,pri}$, $D_{i,sec}$, and $D_{i,ter}$ are dummies for having reached primary, secondary, and tertiary education; and X_i is a vector of controls including gender, an age quartic, and interactions between gender and the age quartic (as in [Lemieux 2006](#); [Autor, Goldin, and Katz 2020](#)).⁷ I restrict the sample to individuals with positive

6. In nearly all countries, the survey was fielded after 2015: see [Online Appendix Table E1](#).

7. Controlling for age rather than potential experience allows the comparison of individuals with more education to those with less education but more work

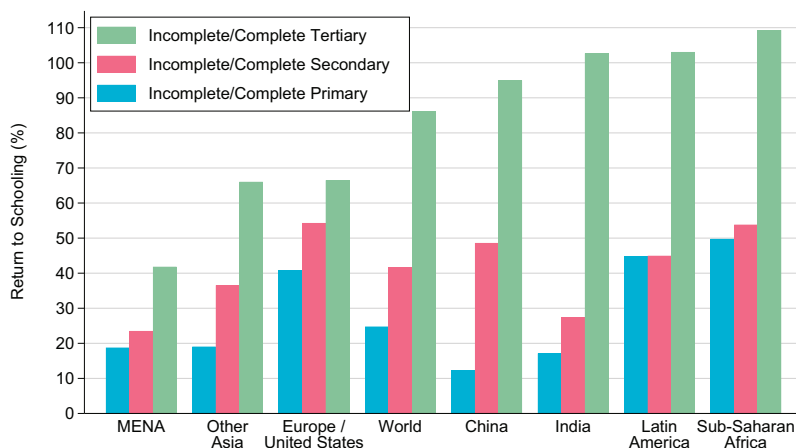


FIGURE II

Total Returns to Schooling by World Region

In the average Sub-Saharan African country, having incomplete or complete primary education increases earnings by 50%, incomplete or complete secondary education further increases earnings by an additional 55%, and incomplete or complete tertiary education further increases earnings by an additional 110%. The return to education level j in a given country is $R_j = 100 \times (\exp(r_j(A^{2019}, L^{2019})) - 1)$, where $r_j(A^{2019}, L^{2019})$ is estimated in each country using equations (11)–(14). Population-weighted averages of returns are estimated in each country. See Online Appendix E1 for additional methodological details.

personal income, including wage earners and self-employed individuals. The returns to reaching each category are:

$$(12) \quad r_{pri}(A^{2019}, L^{2019}) = \ln \left(\frac{w_{pri}^{2019}}{w_{non}^{2019}} \right) = \beta_{pri},$$

$$(13) \quad r_{sec}(A^{2019}, L^{2019}) = \ln \left(\frac{w_{sec}^{2019}}{w_{pri}^{2019}} \right) = \beta_{sec} - \beta_{pri},$$

$$(14) \quad r_{ter}(A^{2019}, L^{2019}) = \ln \left(\frac{w_{ter}^{2019}}{w_{sec}^{2019}} \right) = \beta_{ter} - \beta_{sec}.$$

Figure II plots returns to schooling by world region, converted from log points to percentage changes to ease interpretation. There are large variations across countries and education levels, in line with previous evidence covering a more limited number

experience, which ensures that estimated returns to schooling are net of forgone experience.

of countries (Psacharopoulos 1994; Psacharopoulos and Patrinos 2004; Banerjee and Duflo 2005; Caselli, Ponticelli, and Rossi 2014).⁸ In almost all world regions, returns are higher at higher levels of education. In the average country, the return to having incomplete or complete primary education is 25%, the return to having incomplete or complete secondary education is 40%, and the return to having incomplete or complete tertiary education is 85%.

This suggests that returns to schooling are convex. I do not observe the distribution of years of schooling within each category, however, so it is not possible to conclude that from this database. Importantly, these returns should not be interpreted as returns to completing each level. If the primary education category mostly includes incomplete degrees, for instance, r_{pri} will be lower than the return to completing primary education, and r_{sec} will partly include the effect of finishing primary school.

To shed more light on this issue, I turn to I2D2 (Montenegro and Hirn 2009), a collection of harmonized household surveys maintained by the World Bank, which records exact completed years of schooling for a subset of 62 countries. I investigate the convexity of returns in three ways (see Online Appendix F for more details).

First, I run a piecewise log-linear regression relating the log of personal income to years of schooling, with the same controls as in equation (11) and country fixed effects. Returns to schooling are strongly convex. The return per year of schooling is about 2% for the first 6 years of education, 8% for the next 6 years, and 12% above 12 years (see Table II). This pattern is visible in almost all regions.

Second, I run regressions relating the log of personal income to years of schooling and years of schooling squared.⁹ In every region, the coefficient on years of schooling squared is positive and significant.

Third, I run regressions relating the log of personal income to six dummies for having reached incomplete or complete

8. Variations in returns are generally considered to be driven by the combination of labor supply, labor demand, and education quality differences. Rossi (2022) presents empirical evidence suggesting that education quality only plays a limited role.

9. See Online Appendix Table F4. Online Appendix Table F3 provides results without years of schooling squared.

TABLE II
RETURNS PER YEAR OF SCHOOLING BY LEVEL AND WORLD REGION

	All countries (1)	Latin America (2)	China (3)	India (4)	Other Asia (5)	MENA (6)	Sub-Saharan Africa (7)
Years of schooling: 0–6	0.018*** (0.001)	0.071*** (0.002)	0.064*** (0.010)	0.018*** (0.002)	0.014*** (0.002)	0.017*** (0.002)	0.067*** (0.004)
Years of schooling: 7–12	0.079*** (0.001)	0.090*** (0.001)	0.086*** (0.003)	0.057*** (0.002)	0.072*** (0.002)	0.010*** (0.002)	0.055*** (0.003)
Years of schooling: 13+	0.123*** (0.001)	0.111*** (0.001)	0.109*** (0.004)	0.161*** (0.003)	0.085*** (0.001)	0.042*** (0.001)	0.170*** (0.003)
<i>N</i>	1,291,371	781,614	30,036	122,870	119,659	78,910	158,282
<i>R</i> ²	0.79	0.88	0.21	0.39	0.91	0.74	0.78

Notes. The table reports results of pooled piecewise regressions relating the log of personal income to years of schooling, with knots at 6 and 12 years of education. In India, the return to a year of education is about 2% for the first 6 years, 6% for the next 6 years, and 16% above 12 years. All estimates include country fixed effects and control for gender, an age quartic, and interactions between gender and the age quartic. The full sample covers 62 countries with detailed educational attainment information. Observations are weighted to match the total population of each country. See [Online Appendix E2](#) for more details on the data sources and empirical specification. *** $p < .01$, ** $p < .05$, * $p < .1$.

primary, secondary, or tertiary education. This specification is less straightforward to interpret because the length of degrees varies by education level and across countries. The same pattern of convexity appears nonetheless.¹⁰

Together, these results provide robust evidence that returns to schooling are larger at higher levels of education. This finding contrasts with the previous macroeconomics literature (e.g., [Hall and Jones 1999](#)), which usually assumes concave returns, and is more consistent with recent evidence (e.g., [Rossi 2022](#); [Jedwab et al. 2023](#)). I discuss the reasons underlying this discrepancy with previous work in [Online Appendix F](#).

3. Returns to Schooling: IV Returns. The advantage of OLS returns is that they can be estimated for all 154 countries using a common methodology. The disadvantages are twofold. First, OLS returns may suffer from omitted-variable bias, although this bias has been found to be small ([Card 1999](#); [Gunderson and Oreopoulos 2020](#)). Second, our parameter of interest is the return for those newly educated since 1980, which can differ from the average return in the presence of heterogeneity (e.g., [Heckman, Humphries, and Veramendi 2018](#)).

To make progress in tackling these issues, an alternative option is to use causally identified returns. I have assembled a new collection of instrumental variable (IV) estimates of the returns to schooling. Surveying the literature, I was able to identify 33 cases in which both OLS and IV returns were available for comparison, covering 23 countries or world regions representative of about two-thirds of the world's population.¹¹

[Figure III](#) compares these OLS and IV returns. They are highly correlated ($\hat{\rho} = 0.65$). IV returns almost always exceed or are not significantly different from OLS returns, in line with previous evidence ([Card 1999](#)). The average gap is about 40%. Strikingly, IV returns also appear to be convex. Across studies, the average returns to a year of primary, secondary, and tertiary education are 8.5%, 10%, and 12.5%.

Given the diversity of policies and specifications found across studies, it is difficult to draw conclusions about the exact sources of variations observed in the IV-OLS gap. Nonetheless, I consider

10. See [Online Appendix Table F5](#).

11. See [Online Appendix D.1](#) for more details. [Online Appendix Table D1](#) provides a complete list of these studies.

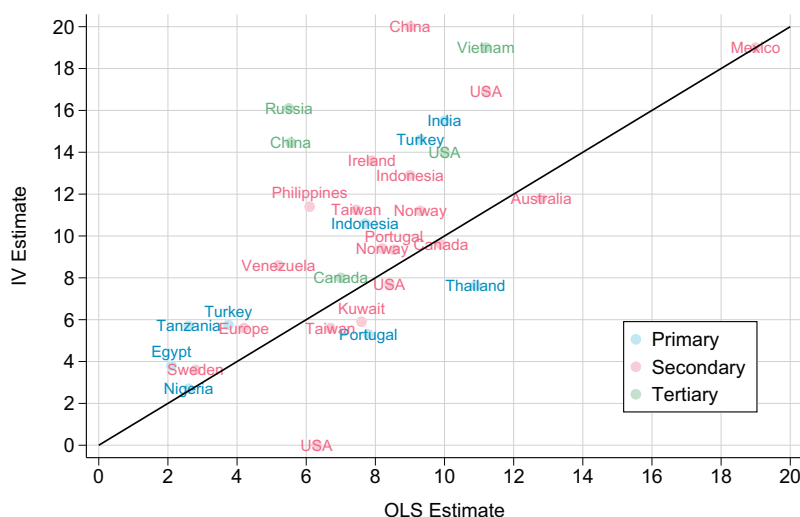


FIGURE III

Returns to Schooling: OLS Versus IV Estimates

Notes: The figure compares OLS and IV estimates of the return to an additional year of schooling. OLS estimates generally correspond to coefficients obtained from a Mincerian equation of the log of earnings on years of schooling, estimated over the entire working-age population. In contrast, IV estimates rely on quasi-experimental variation in access to a specific level of education (primary, secondary, or tertiary). Author's elaboration compiling estimates from existing empirical studies: see [Online Appendix Table D1](#).

a specification in which OLS returns estimated with my data are upward corrected using OLS-IV gaps found across these studies. I compare results using these reduced-form “IV-corrected” returns with those derived from the model.¹²

4. *Educational Attainment Data.* The third input required for the estimation is data on the evolution of educational attainment. The primary source is the [Barro and Lee \(2013\)](#) database, which records the share of individuals with no schooling, primary, secondary, and tertiary education by age and gender in 146 countries since 1950. I complement it with my estimates for missing

12. This reduced-form specification finds support in my analysis of three natural experiments in [Section III.C](#): in each case, the aggregate effect of schooling ends up falling close to the IV return.

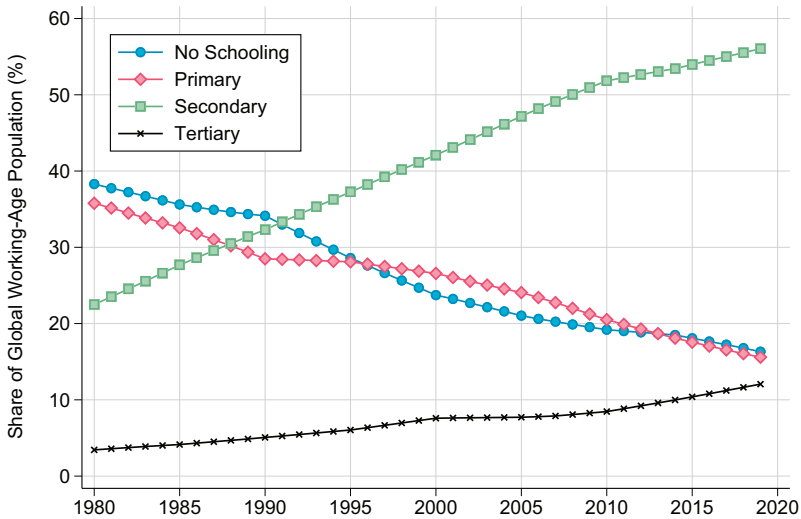


FIGURE IV

Educational Attainment of the World's Working-Age Population, 1980–2019

From 1980 to 2019, the share of the world's working-age population having reached secondary education grew from about 20% to 55%. Author's calculations combining data from the Barro-Lee database and other sources (see [Online Appendix G](#) for more details).

countries, using census data, cohort-level trends observed in the labor force surveys, and other sources (see [Online Appendix G](#)).

[Figure IV](#) plots the evolution of worldwide educational attainment from 1980 to 2019. There has been a rapid expansion of secondary education, from about 20% to 55% of the world's working-age population. This rise was mirrored by a large decline in the share of adults with primary education or no schooling. Tertiary education also expanded significantly, from less than 5% to over 10%.

5. *Global Income Inequality Data.* The last required input is data on the income distribution in each country. I use the World Inequality Database (WID) ([Blanchet et al. 2021](#)), which reports average labor and capital income by percentile in all countries in the world from 1980 to 2019. All components of net national income are allocated to individuals, so that income distributions are consistent with macroeconomic growth rates. I construct estimates of the world distribution of income by converting all

incomes to 2023 PPP US dollars using national income deflators and PPP conversion factors provided in the WID.

III.B. Methodology

I now outline the main elements of the methodology. I first present the steps followed to solve the model. I then discuss how to bring the model to the microdata.

1. Model Specification and Estimation.

i. CES Production Function. Up to now, I worked with two skill groups for illustrative purposes, but the data cover four. To incorporate supply effects on these groups, I introduce nests in the CES production function (following [Goldin and Katz 2007](#); [Fernández and Messina 2018](#)):

$$(15) \quad H^t = \left(A_{ter}^t L_{ter}^t{}^{\frac{\sigma_1-1}{\sigma_1}} + A_{ter}^t L_{ter}^t{}^{\frac{\sigma_1-1}{\sigma_1}} \right)^{\frac{\sigma_1}{\sigma_1-1}},$$

$$(16) \quad L_{ter}^t = \left(A_{sec}^t L_{sec}^t{}^{\frac{\sigma_2-1}{\sigma_2}} + A_{sec}^t L_{sec}^t{}^{\frac{\sigma_2-1}{\sigma_2}} \right)^{\frac{\sigma_2}{\sigma_2-1}},$$

$$(17) \quad L_{sec}^t = \left(A_{non}^t L_{non}^t{}^{\frac{\sigma_3-1}{\sigma_3}} + A_{pri}^t L_{pri}^t{}^{\frac{\sigma_3-1}{\sigma_3}} \right)^{\frac{\sigma_3}{\sigma_3-1}}.$$

At the upper level, output is produced by combining tertiary-educated and non-tertiary-educated workers L_{ter}^t and L_{ter}^t . The intermediate level includes workers with secondary and below-secondary education L_{sec}^t and L_{sec}^t . Finally, the lower level includes workers with primary education L_{pri}^t and no schooling L_{non}^t . Technology parameters are normalized to $A_{ter}^t = 1 - A_{ter}^t$, $A_{sec}^t = 1 - A_{sec}^t$, and $A_{non}^t = 1 - A_{pri}^t$.

ii. Elasticities of Substitution. The second step is to calibrate the elasticities of substitution between skill groups. In my main specification, I use long-run elasticities, which amounts to assuming that firms would adjust their technological mix if education came back to its 1980 level. Two recent studies have made progress in estimating these elasticities. [Hendricks and Schoellman \(2023\)](#) show that wage gains at migration in the United States imply a long-run elasticity ranging from five to eight, depending on the skill cutoff chosen to estimate it (with higher cutoffs implying lower elasticities). [Bils, Kaymak, and Wu \(2022\)](#) estimate that current patterns of worldwide growth, returns to schooling, and cross-country technological differences

imply a long-run elasticity between four and six. In light of this evidence, I specify elasticities of $\sigma_1 = 4$, $\sigma_2 = 6$, and $\sigma_3 = 8$ in my benchmark specification.

iii. Solving the Model. Once the CES production function and elasticities are specified, it is straightforward to solve the model. The returns to schooling are:

$$(18) \quad \ln \left(\frac{w_{pri}^{2019}}{w_{non}^{2019}} \right) = \ln \left(\frac{A_{pri}^{2019}}{A_{non}^{2019}} \right) - \frac{1}{\sigma_3} \ln \left(\frac{L_{pri}^{2019}}{L_{non}^{2019}} \right),$$

$$\ln \left(\frac{w_{sec}^{2019}}{w_{pri}^{2019}} \right) = \ln \left(\frac{A_{sec}^{2019}}{A_{sec}^{2019} A_{pri}^{2019}} \right) - \frac{1}{\sigma_2} \ln \left(\frac{L_{sec}^{2019}}{L_{sec}^{2019}} \right)$$

$$(19) \quad - \frac{1}{\sigma_3} \ln \left(\frac{L_{sec}^{2019}}{L_{pri}^{2019}} \right),$$

$$\ln \left(\frac{w_{ter}^{2019}}{w_{sec}^{2019}} \right) = \ln \left(\frac{A_{ter}^{2019}}{A_{ter}^{2019} A_{sec}^{2019}} \right) - \frac{1}{\sigma_1} \ln \left(\frac{L_{ter}^{2019}}{L_{ter}^{2019}} \right)$$

$$(20) \quad - \frac{1}{\sigma_2} \ln \left(\frac{L_{ter}^{2019}}{L_{sec}^{2019}} \right).$$

These three returns have been estimated using the microdata, and the distribution of educational attainment L^{2019} is observed in the Barro-Lee database. The only unknown parameters are the technology terms A_j^{2019} , which I recover from [equations \(18\)–\(20\)](#). All parameters of the model are then known, and one can directly calculate counterfactual human capital $H(A^{2019}, L^{1980})$ and wages.

iv. Implied Returns to Schooling. What are the returns to schooling implied by this model? In the average country, actual returns to primary, secondary, and tertiary education are about 25%, 40%, and 85%. The counterfactual returns that would prevail absent educational expansion are around 50%, 90%, and 130%.¹³ The model predicts an effect of education on output that is equivalent to using returns falling in between these two bounds. This true return, expressed per year of schooling, can be estimated as:

$$(21) \quad r_{\text{annual}}^*(A^{2019}) = \frac{\ln H(A^{2019}, L^{2019}) - \ln H(A^{2019}, L^{1980})}{S^{2019} - S^{1980}},$$

13. See [Online Appendix Table A1](#).

where S^t is average years of schooling at time t . In other words, the “true” return is the constant return per year of schooling that can reproduce the decline in output predicted by the model. Actual and counterfactual annualized returns can be calculated using the same principle. The actual return is 8%, the true return is 11%, and the counterfactual return is 14% in the average country. Accounting for imperfect substitution is thus equivalent to increasing the return to schooling by about 3 percentage points.

2. *Distributional Growth Accounting.* The next part of the methodology consists in bringing the results of the model to the microdata, which allows estimation of what the income distribution would look like in each country if education had not improved. I construct these counterfactual income distributions in four steps.

i. *Downgrade Education Levels.* The first step is to bring education back to its 1980 level in each country. I implement this “first stage” by sampling individuals and downgrading their education until reaching the counterfactual by age-gender cell (as in [Hershbein, Kearney, and Pardue 2020](#)).

ii. *Adjust Wages.* The second step is to adjust the earnings of the different skill groups. Consider an individual i with initial labor and mixed income $y_{iL}(A^{2019}, L^{2019})$, whose education level is shifted from s_2 to s_1 . Their counterfactual income absent educational expansion is:

$$(22) \quad y_{iL}(A^{2019}, L^{1980}) = \exp \left[\ln(y_{iL}(A^{2019}, L^{2019})) - \ln \left(\frac{w_{s_2}(A^{2019}, L^{2019})}{w_{s_1}(A^{2019}, L^{1980})} \right) \right].$$

For instance, workers downgraded from tertiary to secondary education see their earnings reduced by the 2019 tertiary to counterfactual secondary wage gap. Importantly, workers whose education remains the same also see their wage change. Because of supply effects, the wage of nontreated skilled workers typically increases, whereas that of nontreated unskilled workers typically declines.¹⁴

14. [Online Appendix Table A8](#) illustrates how wages by skill group typically adjust in the average country.

iii. *Derivation of Total Income.* Steps 1 and 2 yield an estimate of how the distribution of labor income would look like if education had not improved. The third step is to move from labor income to total income. In my benchmark specification, I make the conservative assumption that physical capital is not affected by schooling, as in the canonical growth accounting exercise. Counterfactual income of group p then equals $y^p(A^{2019}, L^{1980}) = \exp(y^p(A^{2019}, L^{2019}) - (1 - \alpha^p) \ln \frac{y_L^p(A^{2019}, L^{2019})}{y_L^p(A^{2019}, L^{1980})})$, with α^p its capital income share.

iv. *Growth Accounting.* The fourth and final step is to calculate the share of growth accounted for by education for income group p , as in [equation \(5\)](#):

$$(23) \quad \text{Share}_L^p = \frac{\ln y^p(A^{2019}, L^{2019}) - \ln y^p(A^{2019}, L^{1980})}{\ln y^p(A^{2019}, L^{2019}) - \ln y^p(A^{1980}, L^{1980})}.$$

III.C. Validation from Three Natural Experiments

The ability of this simple model to capture the effect of education may naturally be questioned. My analysis could underestimate the benefits of education if there are human capital externalities (e.g., [Gennaioli et al. 2013](#); [Chauvin et al. 2017](#)) or overestimate them in the presence of signaling effects or negative selection (e.g., [Spence 1973](#); [Fujimoto, Lagakos, and VanVuren 2023](#)). To assess the validity of my framework, I turn to causal evidence from three natural experiments (see [Online Appendix C](#) for more details).

1. *Contexts, Data, and Methodology.* I study three large-scale schooling initiatives: India's District Primary Education Program (1990s–2000s), Indonesia's INPRES school construction program (1970s), and U.S. state compulsory schooling laws (1870s–1960s). Existing work has focused on individual returns to schooling, human capital externalities, and general equilibrium effects of educational expansion ([Khanna 2023](#); [Duflo 2001, 2004](#); [Acemoglu and Angrist 2000](#)). Less is known of the overall effects of these policies on aggregate growth and inequality.

Combining data from existing studies and additional sources, I exploit these natural experiments to estimate the causal effect of educational expansion on growth and inequality across subnational regions. I run variants of the following specification:

$$(24) \quad \ln y_{rt}^i = \gamma_0^i + \gamma_1^i S_{rt} + X_{rt} \beta + \delta_r + \delta_t + \varepsilon_{rt},$$

$$(25) \quad S_{rt} = \alpha_0 + \alpha_1 Z_{rt} + \eta_{rt},$$

where y_{rt}^i denotes the average income of income group i in subnational region r at time t . The objective is to estimate the impact of S_{rt} , the average years of schooling of the working-age population. X_{rt} is a vector of controls, δ_r are subnational region fixed effects, and δ_t are time fixed effects. The parameters of interest are γ_1^i , providing reduced-form estimates of the macroeconomic return to schooling for different groups i .

S_{rt} is instrumented with Z_{rt} , a variable capturing quasi-experimental variation in program exposure. In India, I rely on [Khanna \(2023\)](#), who estimates the impact of the DPP using a regression discontinuity design around the cutoff district literacy rate used to allocate the program. In Indonesia, I instrument district average years of schooling by the number of schools built under the INPRES program, following [Duflo \(2001\)](#). In the United States, the instrument is average required years of schooling across cohorts born in different states ([Acemoglu and Angrist 2000](#); [Guo, Roys, and Seshadri 2024](#)). I then compare the estimated aggregate and distributional effects of each program with those predicted by the model.

2. Main Results. [Figure V](#) plots the main results, comparing estimated and simulated effects of education on average incomes by quintile. All three policies strongly reduced inequality. In India, for instance, one additional year of schooling in a treated district increases the average income of the bottom quintile in this district by 20%, compared with a null effect on that of the top 20%. Aggregate effects of education on earnings range from 8% to 15% (as shown by the dashed line in each figure), comparable to individual returns found in the same contexts ([Duflo 2001](#); [Clay, Lingwall, and Stephens 2021](#); [Khanna 2023](#); [Li 2024](#)).

The model performs remarkably well. In all three cases, simulations predict higher returns to educational expansion at the bottom of the distribution. If anything, the model underestimates benefits for low-income earners in the United States. These results provide reassuring evidence that the methodology developed in this article delivers a good approximation of the distributional effects of educational expansion and may even provide a lower bound on benefits at the bottom of the income distribution.

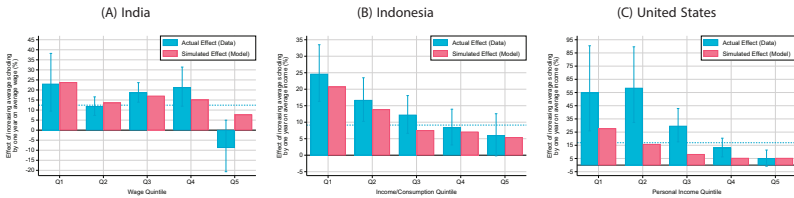


FIGURE V

Validation: Actual Versus Simulated Distributional Effects of Education Expansion Policies in India, Indonesia, and the United States

The three panels compare actual causal effects of educational expansion on the average income of each quintile with simulated effects predicted by the model. Capped spikes correspond to 95% confidence intervals. The dashed line shows the estimated effect of average regional years of schooling on the average income. India: the effect of increasing average district schooling by one year on individual wages, instrumenting schooling with exposure to the District Primary Education Program. Estimates combine NES microdata with exposure to the policy from [Khanna \(2023\)](#). Indonesia: the effect of increasing average district schooling by one year on per adult consumption, instrumenting schooling with exposure to the INPRES program. Estimates combine SUSENAS 1993–2019 microdata with INPRES program intensity from [Duflo \(2001\)](#). United States: the effect of increasing average state schooling by one year on personal income, instrumenting schooling with state compulsory schooling laws. Estimates combine 1940–2000 census microdata with compulsory schooling laws from [Acemoglu and Angrist \(2000\)](#) and [Clay, Lingwall, and Stephens \(2021\)](#). Simulated effects: the return to schooling is set to 16% in India, 11% in Indonesia, and 12% in the United States, following estimates of the individual returns to schooling found in existing studies; the elasticity of substitution is set to six in all three cases. See [Online Appendix C](#) for more details.

IV. EDUCATION AND THE WORLD DISTRIBUTION OF INCOME, 1980–2019

This section presents the main results on the total contribution of education to global economic growth and its distribution from 1980 to 2019. [Section IV.A](#) presents the key takeaways of the distributional growth accounting exercise. [Section IV.B](#) compares these findings to those of a standard growth decomposition. [Section IV.C](#) investigates the sensitivity of the results to alternative assumptions and specifications. [Section IV.D](#) explores heterogeneity across countries, cohorts, time periods, and skill groups.

IV.A. Main Results

I start by presenting the main results of the distributional growth accounting decomposition for the world as a whole in [Table III](#). From 1980 to 2019, global average income per capita increased at an annual rate of 1.6%. The contribution of education

TABLE III
DISTRIBUTIONAL GROWTH ACCOUNTING, WORLD, 1980–2019

	Annual income growth (%)	Contribution of education (percentage points)	Share of growth explained (%)
Full population	1.6	0.7	45
Bottom 50%	2.4	1.1	43
Bottom 20%	1.9	1.1	58
Next 30%	2.6	1.0	41
Middle 40%	1.5	1.1	69
Top 10%	1.6	0.5	30
Top 1%	2.0	0.2	9.9
Top 0.1%	2.5	0.07	2.9

Notes. The table reports actual real annual income growth rates, the contribution of education to growth, and the corresponding share of growth explained by education for different groups of the world distribution of income. The average income of the world's poorest 20% grew by 1.9% per year from 1980 to 2019, 58% of which can be accounted for by education. See [Online Appendix B](#) for additional methodological details.

was 0.7 percentage points per year. Private returns to schooling therefore account for 45% of worldwide per capita income growth over this period.

The contribution of schooling to growth varies significantly by global income group. Growth has been higher for the world's poorest 50%, but benefits from educational expansion have also been higher for this group, so that the share of growth explained by education exceeds 40%. Overall, education can account for 40%–70% of growth for all groups in the global bottom 90%. This share is highest for the bottom 20% (58%) and middle 40% (69%), two groups that have witnessed lower growth and large gains from schooling. The contribution of education is lowest at the very top of the distribution, mainly because the bulk of top incomes consists of capital income, which by assumption is not affected by schooling.

[Figure I](#) provides a more detailed breakdown of the distribution of global economic growth from 1980 to 2019. All individuals in the world are ranked from the poorest 1% to the richest 0.01%. Annual pretax income growth is then calculated for each percentile, together with the contribution of education (lower shaded area) and residual growth from other factors (upper shaded area). Real income gains have been greatest at the middle and very top of the global income distribution, generating what has often been called the “elephant curve” of global inequality and growth ([Lakner and Milanovic 2016](#)). This pattern reflects the conjunction of trends in inequality between and within countries,

TABLE IV
EDUCATION AND GLOBAL POVERTY REDUCTION, 1980–2019

	1980 (%)	2019 (%)	Difference (%)	Share of decline explained (%)
Global poverty: \$2.15/day				
Actual	47	14	–71	
Counterfactual	47	26	–46	35
Global poverty: \$3.65/day				
Actual	61	28	–54	
Counterfactual	61	44	–28	49
Global poverty: \$6.85/day				
Actual	70	49	–30	
Counterfactual	70	63	–10	67

Notes. The table compares the actual evolution of the global poverty headcount ratio to the evolution it would have followed absent educational expansion. Global poverty at \$2.15 per day declined by 71% from 1980 to 2019. Absent educational expansion, it would have declined by 46%. Education thus accounts for 35% of global poverty reduction during this period. All global poverty headcount ratios are calculated using 2017 PPP US\$. The income concept is per capita pretax income, estimated by combining disposable income and consumption distributions from the World Bank data portal with estimates of the distribution of direct taxes and government transfers from Gethin (2024) and Fisher-Post and Gethin (2023). See [Online Appendix Table A3](#) for comparable results using pretax income distributions from the World Inequality Database. [Online Appendix B](#) provides additional methodological details.

including the rise of China and India (middle of the distribution), sluggish economic growth in low-income countries (bottom of the distribution), weak income gains for most households in high-income countries (upper middle of the distribution), and skyrocketing top income inequality in many parts of the world (top end of the distribution). The main contribution of this article is to isolate gains from education, represented by the lower shaded area. These gains have been large, exceeding 1 percentage point for most percentiles in the bottom 90%.

Another indicator that has received much attention is the share of the world's population living in extreme poverty. A difficulty is that poverty headcount ratios are based on counting individuals with incomes below a certain number rather than on actual growth rates, which makes the calculation less conceptually meaningful and more sensitive to the choice of a specific threshold. With these limitations in mind, I extend the decomposition to global poverty rates in [Table IV](#). For this analysis, I use World Bank data, which is the most commonly used data source to measure extreme poverty. Education can account for about 35% of global poverty reduction at \$2.15 a day, 49% at \$3.65 a day, and

TABLE V
FROM STANDARD TO DISTRIBUTIONAL GROWTH ACCOUNTING

Step		Share of growth explained (%), 1980–2019	
		Global average	Global bottom 20%
1	Standard growth accounting	33	16
	Cross-country data, 10% return		
2	+ Adjusted income concept	42	25
3	+ Within-country inequality	42	36
4	+ Heterogeneous labor shares	42	47
5	+ Microdata	31	32
6	+ Supply effects: distributional component	31	46
7	+ Supply effects: aggregate component	45	58

Notes. The table reports the share of global economic growth and growth among the world's poorest 20% explained by education depending on methodological assumptions and data sources used. *Standard growth accounting:* only rely on cross-country data, assuming a uniform 10% return to schooling and that education only affects a fraction of mixed income in each country. *Adjusted income concept:* account for the fact that education affects all of mixed income. *Within-country inequality:* account for within-country inequality—the world's poorest 20% do not all live in the poorest 20% countries. *Heterogeneous labor shares:* account for the fact that labor income is concentrated at the bottom of the distribution in each country. *Microdata:* bring in the microdata, allowing returns to schooling to vary by country and education level. *Supply effects, distributional component:* account for the distributional implications of supply effects—absent educational expansion, the relative wage of skilled workers would be higher. *Supply effects, aggregate component:* account for the aggregate implications of supply effects—absent educational expansion, the return to schooling would be higher. See [Online Appendix B](#) for additional details on each methodological step. [Online Appendices E–G](#) provide additional information on the underlying data sources.

67% at \$6.85 a day—the three thresholds used by the World Bank to measure global poverty.¹⁵

IV.B. Comparison with Standard Growth Accounting

A first way of better understanding these estimates is to study the effect of applying each of the estimation steps outlined in [Section III](#). This analysis allows comparison of my results to those of a canonical growth accounting decomposition, which is useful to isolate the different channels driving the results. [Table V](#) displays the share of global average economic growth and global bottom 20% income growth explained by education with different data sources and assumptions.

1. *The Standard Growth Accounting Decomposition.* I start by presenting results from a standard growth accounting decom-

15. [Online Appendix Table A3](#) reproduces this analysis using the WID. [Online Appendix Table A2](#) reproduces [Table III](#) with World Bank data.

position. I follow [Barro and Lee \(2015\)](#), who estimate the fraction of global economic growth explained by human capital from 1960 to 2010. This decomposition only requires three ingredients: per capita net national income data (taken from the WID), capital income shares α (taken from the Penn World Tables to follow [Barro and Lee 2015](#)), and an estimate of the Mincerian return to schooling (set at $r = 10\%$ per year). Human capital at time t is $H^{t, \text{standard}} = e^{rS^t}$, where S^t are average years of schooling of the working-age population. The contribution of education to growth is thus:

$$(26) \quad \text{Share}_L^{\text{standard}} = (1 - \alpha) \frac{r \times (S^{2019} - S^{1980})}{\ln Y^{2019} - \ln Y^{1980}}.$$

The first row of [Table V](#) presents the results. Education accounts for 33% of global average economic growth. The second column shows results for the global bottom 20%. Because this growth accounting decomposition relies on cross-country data, the poorest 20% have to be defined as the poorest 20% of countries (population weighted). Educational progress has been relatively weak in these countries. As a result, education explains about 16% of growth for the global bottom 20%.

2. Adjusting the Income Concept. A first problem with this approach is that capital income shares in the Penn World Tables include most of mixed income. As a result, the implicit assumption in [equation \(26\)](#) is that the return to schooling only applies to a small fraction of mixed income. This does not appear to be true. OLS returns estimated in [Section III.A](#) include all of mixed income and are, on average, indistinguishable from those estimated on wages only.¹⁶ Aggregate effects of schooling estimated using the natural experiments studied in [Section III.C](#) also include mixed income. The income concept used should thus include all of mixed income, because this is the concept returns to schooling estimated in this study apply to.

The second row of [Table V](#) presents the results when the return to schooling applies to all of mixed income. The Penn World Tables do not provide this decomposition, so I turn to the factor income shares estimated by [Bachas et al. \(2022\)](#). The contribution

16. See [Online Appendix Table F1](#). Let us note that this result need not imply that the capital of self-employed workers is unproductive. Rather, it suggests that education enables these workers to accumulate more capital or to make this capital more productive.

of education to average growth rises to 42%, and its contribution to bottom 20% growth to 25%. In the [Online Appendix](#), I investigate the sensitivity of my results to a more modest adjustment in which only 75% of mixed income is affected by education.¹⁷

3. *Incorporating Within-Country Inequality.* In a third step, I account for within-country inequality: the global poorest 20% no longer correspond to the poorest 20% of countries. All other methodological ingredients stick to the standard growth accounting exercise. The average income of each income group is reduced by the same proportion in each country.

By construction, accounting for within-country inequality leaves the share of aggregate growth explained unchanged. However, it raises the contribution of education to bottom 20% growth from 25% to 36% for two main reasons. First, the bottom 20% is now a mix of individuals from low-income and middle-income countries, some of which witnessed significant educational progress. Second, inequality has risen rapidly in many countries. This second factor increases the contribution of education simply because there is less growth among the global poor to be explained than what cross-country data suggest.

4. *Incorporating Within-Country Capital Income Concentration.* Fourth, I account for the fact that capital income is concentrated at the top of the distribution in each country. For the majority of individuals belonging to the bottom 90%, almost all of income consists of wages or mixed income. The passthrough from schooling to income is thus close to 100% for low-income earners, rather than equal to the aggregate labor income share. As in the third step, this does not affect the share of aggregate growth explained by education. However, it raises the contribution of education to bottom 20% growth significantly, from 36% to 47%.

17. See [Online Appendix Table A4](#). This adjustment is equivalent to using the benchmark labor shares estimated in [Bachas et al. \(2022\)](#), who attribute 75% of mixed income to labor as is common in the literature. The share of global economic growth explained is almost identical, highlighting that the difference between rows (1) and (2) primarily comes from differences between the Penn World Tables and [Bachas et al. \(2022\)](#) databases. The capital share is almost 50% in the Penn World Tables in the average country, likely reflecting a choice to attribute most of mixed income to capital, compared to 33% in [Bachas et al. \(2022\)](#). The share of global bottom 20% growth explained declines more due to the larger share of mixed income in total income in low- and middle-income countries.

5. *Bringing in the Microdata.* Fifth, I bring in the micro-database. The main difference is that returns to schooling are now allowed to vary by country and education level. The contribution of education to average growth declines, mainly because the implied average Mincerian return in my data is closer to 8% than 10%. The contribution of education to global bottom 20% growth declines even more, reflecting the fact that the world's poorest individuals have mostly benefited from expansions in primary and secondary education, whose average annual returns are even lower than 8% (see [Table II](#)).

6. *Accounting for Supply Effects: Distributional Component.* Sixth, I account for the distributional component of supply effects: education reduces inequality by increasing the relative supply of skilled workers. This step of the methodology requires data on the joint distribution of income and education, so it can only be conducted with the microdatabase.

To isolate this channel, I only adjust relative wages, leaving the average income unchanged in each country. The contribution of education to average growth is therefore the same as in the previous step. The share of growth explained by education among the world's poorest 20% increases from 32% to 46%, reflecting the large effect of education on inequality within countries.

7. *Accounting for Supply Effects: Aggregate Component.* Finally, I account for the aggregate component of supply effects: not expanding education would have been more detrimental to growth than what returns to schooling observed in 2019 suggest. This final step affects wages differentially across countries and skill groups, generating both aggregate and distributional effects. The contribution of education to average growth rises from 31% to 45%, and the contribution of education to global bottom 20% growth increases from 46% to 58%, yielding the benchmark estimate presented at the beginning of this section.

8. *Summary.* In summary, the results presented in [Table V](#) tell us two facts about education and global poverty. First, education explains almost four times more growth for the global poor than a standard growth accounting exercise would suggest. Changes in within-country inequality, capital income concentration, differential returns, and general equilibrium effects imply a much more complex picture than that depicted by a standard

TABLE VI
SENSITIVITY TO ALTERNATIVE SPECIFICATIONS

	Share of growth (%) explained, 1980–2019	
	World average	Global bottom 20%
Benchmark specification	45	58
IV returns to schooling	51	71
Alternative degrees of substitutability		
High substitutability: $\sigma_1 = 5, \sigma_2 = 8, \sigma_3 = \infty$	41	47
Low substitutability: $\sigma_1 = 2, \sigma_2 = 4, \sigma_3 = 6$	53	78
Very low substitutability: $\sigma_1 = 1.5, \sigma_2 = 2.5, \sigma_3 = 4$	62	109
Alternative production functions		
CES with alternative nesting structure	44	55
CES with no nest	43	57
CES with imperfect substitution between age groups	44	57
Physical capital affected by education	62	67

Notes. The table reports how results about the share of growth explained by education vary depending on assumptions regarding the returns to schooling, the degree of substitutability between skill groups, functional forms, and whether physical capital is affected by education. See [Online Appendix D](#) for additional details on each specification.

decomposition based on aggregate data. Second, accounting for the distributional effects of schooling is essential. General equilibrium effects account for one-third of the contribution of education to global poverty reduction by redistributing a large share of schooling gains from high-skilled to low-skilled workers.

IV.C. *Sensitivity to Alternative Specifications*

[Table V](#) highlights how assumptions regarding (i) the returns to schooling, (ii) supply effects, and (iii) which types of income are affected by education have important implications for the analysis. I study the sensitivity of my results to alternative assumptions on these three dimensions in [Table VI](#).

1. *Returns to Schooling.* I first investigate the implications of using IV-corrected returns to schooling instead of model-based estimates.¹⁸ This increases the contribution of education to

18. This is equivalent to using IV-corrected returns instead of aggregate effects predicted by the model in step 7 of [Table V](#). I still assume that distributional effects of education are the same as in the benchmark specification. See [Online Appendix D.1](#) for more details.

average growth to 51% and its contribution to growth for the world's poorest 20% to 71%.

2. *Elasticities of Substitution.* A second important assumption relates to the degree of substitutability between skill groups. My benchmark specification assumes $\sigma_1 = 4$, $\sigma_2 = 6$, and $\sigma_3 = 8$. Table VI reports results from three alternative specifications. In the high substitutability scenario, I assume elasticities at the upper end of those found in the literature (Hendricks and Schoellman 2023). In the low substitutability scenario, I set long-run elasticities closer to the lower bound of four estimated in Bils, Kaymak, and Wu (2022). In the very low substitutability scenario, I use short-run elasticities found in the literature (Katz and Murphy 1992; Fernández and Messina 2018; Khanna 2023), which amounts to assuming that bringing education back to its 1980 level would lead to no endogenous adjustment in labor demand.

The share of average growth explained is moderately affected by these alternative assumptions, ranging from 41% to 62%. The share of global bottom 20% growth explained is more sensitive, ranging from 47% in the high substitutability scenario to over 100% in the very low substitutability scenario. In a world in which labor demand would not adjust, not expanding education would have led to huge increases in within-country inequality, implying no income gains for the global poor.

3. *Production Function Specification.* I also investigate the sensitivity to using three alternative specifications of the CES production function (see Online Appendix D.2 for more details). The first one assumes that firms first choose between workers with below- and above-secondary education, and then choose between workers in each of these two groups. The second one is a standard CES production function with no nest and a single elasticity. The third one further accounts for imperfect substitution between age groups and returns to experience, which have been shown to be an important determinant of cross-country income differences (Lagakos et al. 2018; Rossi 2020; Jedwab et al. 2023). As shown in Table VI, the results are essentially identical to those obtained with my benchmark functional form.

4. *Physical Capital Adjustment.* My main results assume that education has no effect on physical capital per person. This

assumption is conservative: one should expect physical capital to endogenously respond to educational expansion. More specifically, [equation \(4\)](#) can be rearranged as:

$$(27) \quad 1 = \frac{\alpha}{1 - \alpha} \frac{\Delta \ln \left(\frac{K}{Y} \right)}{\Delta \ln Y} + \frac{\Delta \ln(z)}{\Delta \ln Y} + \frac{\Delta \ln(H)}{\Delta \ln Y}$$

as is standard in the literature (e.g., [Hsieh and Klenow 2010](#)). In this specification, capital per person is allowed to change in response to human capital accumulation, consistent with a long-run growth model in which human capital and total factor productivity do not affect the capital-output ratio in steady state. The passthrough from education to output is then 1 instead of $1 - \alpha$. In this scenario, education explains 62% of global economic growth and 67% among the world's poorest 20%.¹⁹

5. Other Robustness Checks. Finally, I conduct four additional robustness checks.

A first check focuses on educational attainment categories. In my main specification, I assume that workers are homogeneous within the four skill groups. One may be concerned that the distribution of human capital in these categories (e.g., incomplete versus complete secondary education) may have changed over time. I study this possibility in [Online Appendix D.3](#), focusing on 37 countries with sufficiently high-quality data. The results are insensitive to this added layer of detail.

A second concern relates to who benefits from schooling. In the main specification, I randomly sample individuals and downgrade their education until matching 1980 levels. I investigate the implications of accounting for heterogeneous educational expansion by age, gender, subnational region (India), and race (South Africa, United States) in [Online Appendix D.4](#). Using more refined categories has limited effects.

A third robustness check centers on school attendance: if education had not improved, more adolescents and young adults would be working today instead of being in school. I derive an upper bound on this opportunity cost of schooling in [Online Appendix D.5](#). The main results are barely affected.²⁰

19. [Online Appendix Table D3](#) presents the full distributional growth accounting decomposition by income group with this specification.

20. Another related margin through which education could affect poverty is fertility. Empirical evidence is mixed. Female education has been found to reduce

A last concern relates to education quality. If education quality has changed, then educational attainment in 1980 and 2019 might not be comparable. I investigate this concern at length in [Online Appendix D.6](#). Although data are scarce, available evidence suggests stagnating quality in most countries ([Altinok, Angrist, and Patrinos 2018](#); [Angrist et al. 2021](#); [Le Nestour, Moscoviz, and Sandefur 2022](#)). Even under conservative assumptions on a potential decline in quality, my main results are unaffected.

IV.D. Heterogeneity

I conclude this section with a discussion on heterogeneity, decomposing the worldwide effects of education between and within countries, by world region, and by time period.

1. *Growth Accounting Within and Between Countries.* I first decompose the effect of education on inequality between and within countries.²¹ From 1980 to 2019, global inequality remained constant. Absent educational progress, it would have risen by 20%, primarily because education reduces within-country inequality—the effect of schooling on between-country inequality has been close to zero. Education has thus sufficiently mitigated the rise of within-country inequality to keep overall global inequality stable.

2. *Growth Accounting by World Region.* A second way of better understanding the results is to decompose them by world region. In the average country, education explains about 25% of aggregate growth. This figure is highest in Europe and Northern America, Latin America, and Sub-Saharan Africa, and lowest in China and India. In all regions, education explains over 30% of growth among the poorest 50%.²²

early pregnancy ([Osili and Long 2008](#); [Duflo, Dupas, and Kremer 2015](#); [Ozier 2018](#)) but not necessarily lifetime fertility ([Duflo et al. 2024](#)). The implications for my results are thus unclear. If anything, accounting for negative effects of education on dependency ratios would likely magnify the pro-poor nature of educational expansion.

21. See [Online Appendix Table A5](#), which reports a Theil decomposition of global inequality from 1980 to 2019.

22. See [Online Appendix Table A6](#). I also provide complete growth decompositions by world region in [Online Appendix Table D4](#).

3. *Growth Accounting by Time Period.* Finally, the results can also be decomposed by time period. The share of growth explained by schooling has been lower in recent years, mainly due to the exceptional catchup of China and India, but it has remained high for the global poor. Education accounts for about 20% of aggregate growth since 2000, but 40% among the world's poorest 20%.²³

V. THE ROLE OF SKILL-BIASED TECHNICAL CHANGE

The previous section focused on the total contribution of education, that is, the effect of education given skill-biased technical change. In this section, I turn to isolating the role played by skill-biased technical change in making education valuable in 109 countries from 2000 to 2019. [Section V.A](#) presents the data and methodology. [Sections V.B](#) and [V.C](#) provide estimates of skill-biased technical change and the independent contribution of education to growth. [Section V.D](#) proposes a tentative analysis of the contribution of skill-biased technical change to worldwide growth and poverty reduction.

V.A. *Historical Survey Microdata*

Quantifying the independent contribution of education to growth for the world as a whole over 1980–2019 would require surveys fielded in 1980 in each country. Such surveys do not exist. However, I was able to find surveys fielded around 2000 in a subsample of countries (see [Online Appendix E](#) for more details).

1. *Data Sources.* The main data source is the I2D2 database, a repository of household surveys maintained by the World Bank. The coverage of labor force surveys is lower than in the ILO microdatabase, which is why I do not use it as my primary source in the main analysis. Its historical coverage is much greater, however, which makes it better suited for estimating the independent effect of education since 2000.

I searched for all surveys fielded in the late 1990s or early 2000s with detailed information on personal income and educational attainment. Such surveys are available for 71 countries in I2D2. I complement them with other surveys from country-specific sources and other data portals. The resulting micro-

23. See [Online Appendix Table A7](#).

database covers 109 countries located in all regions and representative of about 80% of the world's population.

The quality of these surveys is highly variable and generally lower than the ones used in the rest of the article. They have lower sample size and are typically not labor force surveys. Sub-Saharan Africa is strongly underrepresented. Furthermore, the distribution of educational attainment is less consistent with the Barro-Lee database. For these reasons, I view the analysis presented in this section as more suggestive.

2. Methodology. The methodology is the same as in the rest of the article, except that the quantity of interest is $H(A^{2000}, L^{2019}) - H(A^{2000}, L^{2000})$. I start by solving the model exactly as in [Section III.B](#). First, I estimate the returns to schooling in 2000, $r_j(A^{2000}, L^{2000})$, using the historical survey micro-database. Second, I use these returns to recover the technology parameters A_j^{2000} . All parameters of the model are then known and one can directly calculate the independent contribution of education to growth. The independent effect of education on inequality is the same as its total effect, as highlighted in [equation \(10\)](#), so it is straightforward to estimate distributional effects of education in each country.

V.B. Skill-Biased Technical Change Since 2000

I start by presenting two sets of estimates of skill-biased technical change, depending on whether one assumes short-run or long-run elasticities.²⁴ Using long-run elasticities should imply rates of exogenous technological change that are close to zero on average, since it is equivalent to assuming that changes in demand for skilled labor result from educational expansion (see [Section II.B](#)). In other words, results with short-run elasticities should be interpreted as reflecting actual skill-biased technical change, while results with long-run elasticities correspond to residual skill-biased technical change when one assumes that changes in labor demand reflect endogenous responses to educational expansion.

[Table VII](#) presents results on skill-biased technical change by world region over 2000–2019, defined as annualized changes

24. [Online Appendix Table A9](#) reports the corresponding returns to schooling by level and world region in 2000 and 2019.

TABLE VII
ANNUAL RATES (%) OF SKILL-BIASED TECHNICAL CHANGE BY WORLD REGION, 2000–2019

	Assuming short-run elasticities			Assuming long-run elasticities		
	Primary $\frac{A_{pri}}{A_{non}}$	Secondary $\frac{A_{sec}}{A_{sec}}$	Tertiary $\frac{A_{ter}}{A_{ter}}$	Primary $\frac{A_{pri}}{A_{non}}$	Secondary $\frac{A_{sec}}{A_{sec}}$	Tertiary $\frac{A_{ter}}{A_{ter}}$
Average country	-0.7	-0.1	+0.8	-1.1	-1.2	-0.4
Europe/Northern America		+1.5	+1.9		+0.6	+0.9
Latin America	-0.6	+1.0	+0.7	-0.7	-0.4	-0.4
China	-1.0	-0.6	-1.7	-1.6	-2.2	-2.6
India	-0.9	-2.1	+1.0	-1.2	-2.9	-0.5
Other Asia-Pacific	-0.4	+1.1	+2.5	-0.8	-0.1	+0.8
Middle East and North Africa	-0.1	+1.3	+1.3	-0.4	+0.2	+0.1
Sub-Saharan Africa	-0.9	+0.5	-0.0	-1.1	-0.3	-0.2

Notes. The table reports annualized rates of skill-biased technical change by world region and in the average country, that is, $\frac{1}{19}(\ln \frac{A_t^{2019}}{A_t^{2000}} - \ln \frac{A_t^{2000}}{A_t^{2000}})$ for pairs of skill groups j and i in each country. Subscripts non , pri , sec , ter , and ter refer to workers with no schooling, primary, below secondary, secondary, below tertiary, and tertiary education, respectively. Short-run elasticities: $\sigma_1 = 1.5$, $\sigma_2 = 2.5$, $\sigma_3 = 4$. Long-run elasticities: $\sigma_1 = 4$, $\sigma_2 = 6$, $\sigma_3 = 8$. Population-weighted averages of technology terms are estimated in each country. The table covers 109 countries with data on incomes and education around 2000 and 2019. Estimates are reported for countries with sample sizes sufficiently large to estimate changes in relative demand for primary (47 countries), secondary (83 countries), and tertiary education (107 countries). See [Online Appendices B and E](#) for additional details on estimation methods and data sources.

in relative demand for primary-educated workers $\frac{A_{pri}}{A_{non}}$, secondary-educated workers $\frac{A_{sec}}{A_{sec}}$, and tertiary-educated workers $\frac{A_{ter}}{A_{ter}}$. Focusing on short-run elasticities, demand for primary-educated workers is found to have declined, demand for secondary-educated workers to have stagnated, and demand for tertiary-educated workers to have slightly increased in the average country. There is considerable heterogeneity across world regions. Demand for skilled workers has declined in Sub-Saharan Africa, India, and especially China, mirroring the large drop in the return to college documented by [Hanushek, Wang, and Zhang \(2023\)](#). Skill-biased technical change has also been limited in Latin America, confirming previous evidence ([Fernández and Messina 2018](#)). Europe and Northern America, where demand for college-educated workers has increased rapidly, stand out as the exception.

The results are similar with long-run elasticities, except that the exogenous component of skill-biased technical change is found to have been zero or negative in most regions. In other words, almost all skill-biased technical change is interpreted as having resulted from educational expansion if one uses long-run elasticities. This should not come as a surprise, given that long-run elasticities are precisely calibrated to rationalize variations in skilled labor supply and wage premia without invoking technical change.

V.C. The Contribution of Education to Growth with and Without Skill-Biased Technical Change

I now turn to studying the role played by skill-biased technical change in amplifying the benefits of schooling. [Table VIII](#) compares the independent (without skill-biased technical change) and total (with skill-biased technical change) shares of aggregate growth explained by education by world region. As discussed in [Section IV.D](#), the total contribution of education is lower over 2000–2019 than 1980–2019, mainly because of the catchup of China and India. In the average country, it reaches 20%–25%.

Skill-biased technical change has significantly amplified the effect of education (columns (1)–(3)). In the average country, the independent contribution of schooling is about 70% of its total effect: 30% of the benefits of education have been made possible by growing demand for skilled labor. There are large variations across regions. In Europe and Northern America, the interaction between education and technology amounts to more than half of the total contribution of education. In China and India, this

TABLE VIII
SHARE OF GROWTH (%) EXPLAINED BY EDUCATION BY WORLD REGION WITH AND WITHOUT SKILL-BIASED TECHNICAL CHANGE,
2000–2019

	Assuming short-run elasticities			Assuming long-run elasticities		
	Without skill-biased technical change (1)	With skill-biased technical change (2)	Without / with (1) / (2) (3)	Without skill-biased technical change (4)	With skill-biased technical change (5)	Without / with (4) / (5) (6)
Average country	18	25	72	22	21	102
Europe/Northern America	16	36	44	23	33	72
Latin America	53	64	82	59	57	103
China	9	8	115	10	7	152
India	18	15	122	20	12	161
Other Asia-Pacific	14	29	48	20	22	88
Middle East and North Africa	9	15	60	11	11	102
Sub-Saharan Africa	17	23	75	21	21	99

Notes. The table compares the independent and total shares of growth (in percent) explained by education in the average country and by world region. Columns (1) and (4) report the “independent” effect that education would have had on growth absent skill-biased technical change. Columns (2) and (5) report the “total” effect that education had on growth given skill-biased technical change. Short-run elasticities: $\omega_1 = 1.5$, $\omega_2 = 2.5$, $\omega_3 = 4$. Long-run elasticities: $\sigma_1 = 4$, $\sigma_2 = 6$, $\sigma_3 = 8$. Population-weighted averages of shares of growth explained in each country. The table covers 109 countries with data on incomes and education around 2000. See [Online Appendices B and E](#) for additional details on estimation methods and data sources.

TABLE IX
SHARE (%) OF WORLDWIDE GROWTH EXPLAINED BY EDUCATION BY INCOME
GROUP WITH AND WITHOUT SKILL-BIASED TECHNICAL CHANGE, 2000–2019

	Assuming short-run elasticities			Assuming long-run elasticities		
	Without skill-biased technical change (1)	With skill-biased technical change (2)	Without / with (1) / (2) (3)	Without skill-biased technical change (4)	With skill-biased technical change (5)	Without / with (4) / (5) (6)
Full population	17	25	68	21	21	98
Bottom 50%	48	51	95	36	32	114
Bottom 20%	63	66	95	44	39	112
Next 30%	46	48	94	35	30	115
Middle 40%	22	27	82	23	21	109
Top 10%	1.5	15	10	13	18	74

Notes. The table compares the independent and total shares (in percent) of growth explained by education for different groups of the world distribution of income. Columns (1) and (4) report the “independent” effect that education would have had on growth absent skill-biased technical change. Columns (2) and (5) report the “total” effect that education had on growth given skill-biased technical change. Short-run elasticities: $\sigma_1 = 1.5$, $\sigma_2 = 2.5$, $\sigma_3 = 4$. Long-run elasticities: $\sigma_1 = 4$, $\sigma_2 = 6$, $\sigma_3 = 8$. The estimation is based on data for 109 countries with information on incomes and education around 2000, which together are representative of about 80% of the world’s population. See [Online Appendices B](#) and [E](#) for additional details on estimation methods and data sources.

interaction is negative: education would have had larger effects if labor demand had remained constant.

With long-run elasticities, the gap between the independent and total contributions of education is much smaller—about zero in the average country and below 15% in all regions but Europe and North America. In other words, exogenous skill-biased technical change is found to have played almost no role if one assumes that labor demand reflects an endogenous response of firms to educational expansion.

V.D. Skill-Biased Technical Change and the World Distribution of Income Since 2000

I conclude with a tentative extension of this analysis to the world distribution of income. I restrict the sample to the 109 countries with available data in both 2000 and 2019. I then construct measures of the independent and total contributions of education to growth for different global income groups, assuming that these 109 countries are representative of the world.

[Table IX](#) presents the results for short-run and long-run elasticity specifications. As discussed previously, using lower

elasticities implies that the total effect of education is greater, especially among the global poor, but also that a larger fraction of this effect results from an interaction between education and technology.

With short-run elasticities, the total contribution of education to worldwide growth reaches 25%, about 30% of which was made possible by skill-biased technical change. This figure is much larger at the middle and top ends of the world distribution of income, reflecting the rapid rise in demand for skilled labor in high-income countries. For the global poor, in contrast, technological change has only played a modest role. Education accounts for 66% of growth among the world's poorest 20%, and would still have accounted for 63% of growth absent skill-biased technical change, reflecting the low or negative growth in skilled labor demand in China, India, and Sub-Saharan Africa documented in [Table VII](#). The results are similar with long-run elasticities, except that the total contribution of education is estimated to have been smaller (30%–35% for the world's poorest 20%) and labor demand dynamics are estimated to have played a smaller role.

VI. CONCLUSION

This article quantified the role of education in the historical reduction of global poverty. Combining macroeconomic growth decomposition with the canonical model of education and the wage structure, I proposed a distributional growth accounting framework identifying the contribution of education to growth within and across countries. Under conservative assumptions, education can explain a large fraction of income gains among the world's poorest individuals, on the order of 60%–70% and potentially more. This finding puts public education policies at the center of the remarkable reduction of global poverty observed in the past decades.

My focus was on the poverty-reducing effects of education from an accounting perspective, but much remains to be understood of the drivers of educational expansion itself. On the one hand, one should expect schooling decisions to respond to the economic environment, whether it is rising demand for skilled labor or other factors (e.g., [Foster and Rosenzweig 1996](#); [Restuccia and Vandenbroucke 2013](#)). On the other hand, there are many examples of significant and long-lasting disconnections between the two: for instance, growth in the supply of skilled work-

ers has stalled in the United States despite rapid skill-biased technical change (Autor, Goldin, and Katz 2020), while the exact opposite has happened in Latin America (Fernández and Messina 2018). Many other factors could also explain educational progress, with public policies being a natural candidate (Bharti and Yang 2024; Gethin 2024). Identifying the respective roles of technological progress, behavioral responses, and institutions in driving education and its effects on long-run growth remains an important avenue for future research.

More generally, the results presented in this article illustrate how combining cross-country microdata with simple conceptual frameworks can help uncover new insights on the long-run drivers of the world distribution of income that would otherwise be missed by more aggregate analyses. Similar methods could be developed to study other key transformations, such as trade globalization, structural change, financial integration, and international migration. The microeconomics literature provides ample and growing empirical evidence on the economic effects of these factors in specific contexts. Combined with new microdata similar to those used in this study, complementary data sources, and adequate theoretical frameworks, this evidence could be aggregated to shed light on the role played by these long-run processes in the reduction of global poverty and inequality.

SUPPLEMENTARY MATERIAL

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

DATA AVAILABILITY

The data underlying this article are available in the Harvard Dataverse, <https://doi.org/10.7910/DVN/5CSZUS> (Gethin 2025).

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